Kunal Bajaj

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Experience

Google May 2026 – August 2026

Software Engineer Intern, AI Safety *Kitchener, ON*

- Extended a **C++** inference path for production AI workloads across Google Workspace, supporting **8K+ combined queries per second** at peak.
- Migrated **3 production safety classifiers** from a deprecated C++ service to a multi-agent LLM critic platform, replacing custom service code with **declarative configuration** and production orchestration.
- Benchmarked 3 Gemini model sizes and sampling configurations on an internal serving harness, reducing TPU footprint by **30%** while maintaining **82.5% precision** and flat cost and latency at full traffic.

PointClickCare January 2026 – April 2026

Associate Applied Researcher Intern (AI/ML) *Mississauga, ON*

- Owned AutoMed from **design document to production**: an agent that proposes, tests and verifies medical NER policies in parallel, improving recall from **0.72 to 0.94** and precision from **0.81 to 0.95** without fine-tuning.
- Designed a human-versus-LLM evaluation framework; iterative chunking and re-labeling raised entity-extraction **F1 from 0.78 to 0.89** and reduced annotation inconsistencies by **40%**.
- Built a synthetic-document pipeline grounded in medical reference material, generating **50K+ samples** and expanding document coverage by **35%** for fine-tuning and evaluation.

Sheridan Centre for Applied AI Oakville, ON

Applied AI Engineer Intern *May 2025 – December 2025*

- Shipped a **FastAPI and React** RAG chatbot for 200+ PDF uploads; combined BM25 and dense-embedding retrieval to reach **95% retrieval accuracy**.
- Fine-tuned language models with **LoRA** for VAC's **trauma-informed** veteran-support chatbot.
- Built an **LLM-as-a-judge** evaluation harness that cut manual review effort by **60%** and improved output consistency across agent releases.
- Hosted **Qwen and Llama** with **vLLM** for applied-AI research projects and optimized inference serving.

AI and Process Optimization Analyst *September 2024 – April 2025*

- Forecast 700+ NICU bed allocations at 90% accuracy with grid-searched **XGBoost** models; delivered predictions to a healthcare client through an **internal production API**.

AI Research Assistant *October 2023 – September 2024*

- Delivered a LangChain, Pinecone and FastAPI RAG assistant for Oakville Public Library, reaching 82% accuracy on service inquiries during a **3-month pilot** with **25+ patrons**.

Open Source Contributions

LangChain | *2 Merged PRs* · **Streamlit** | *2 Merged PRs*

- Contributed **4 merged PRs** to LangChain and Streamlit: configurable parameters, cache and error-handling improvements, and a new Streamlit widget.

Projects

Coda – 3D Conferencing & AI Recap Engine | *GitHub* | *Live Demo*

- Designed a Python multi-agent and **tool-use engine** for real-time meeting recaps with prompt chaining, Playwright and async FastAPI, reducing synthesis latency by **35%**.
- Implemented separate async workers for audio, vision and summaries, with **bounded queues**, explicit overload errors and timed shutdown; chained primary, secondary and **fallback summarizers**.

Education

Sheridan College September 2022 – December 2026

Honours Bachelor of Computer Science (Mobile Computing) *Oakville, ON*

Technical Skills

Languages: Python, TypeScript, C++, Swift, SQL
AI Engineering: LangGraph, LangChain, RAG, Tool Use, LLM Evaluation, OpenAI API, vLLM, LoRA
Delivery: FastAPI, React, PostgreSQL, Docker, Azure
Core: Data structures & algorithms, System design, Git, Code review, Debugging